

but on the extent to which they result in a permanent sink for NO_x . Such mechanistic information can be obtained only from direct studies of the disappearance rates of NO_3 (and N_2O_5) and a characterization of reaction products.

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Specific Rate Constants $k(E, J)$ for the Unimolecular Dissociations of H_2CO and D_2CO

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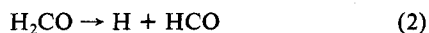
Statistical calculations of J - and E -dependent specific rate constants $k(E, J)$ for the unimolecular complex eliminations $\text{H}_2\text{CO} \rightarrow \text{H}_2 + \text{CO}$ and $\text{D}_2\text{CO} \rightarrow \text{D}_2 + \text{CO}$ and the unimolecular simple bond fissions $\text{H}_2\text{CO} \rightarrow \text{H} + \text{HCO}$ and $\text{D}_2\text{CO} \rightarrow \text{D} + \text{DCO}$ are presented. RRKM type calculations including the K rotor with proper constraints were applied to the elimination reactions, whereas a simplified statistical adiabatic channel calculation was applied to the bond fission reactions. The J dependences of the specific rate constants are demonstrated in detail. The calculations show "channel switching" for rotationally hot molecules. Representations for $k(E, J)$, including tunneling contributions, are given.

Introduction

In the dissociation of formaldehyde in its electronic ground state two basically different reaction channels with very close threshold energies compete, the molecular elimination reaction leading to molecular products



and the simple bond fission reaction leading to radical products



The two-channel character of the dissociation has been recognized in thermal dissociation experiments only recently.¹ Earlier results² were attributed to a single channel only.

Detailed experiments on the photochemistry of formaldehyde more clearly show the competition between two channels in the threshold range. These experiments as well as highly resolved measurements of fluorescence lifetimes (for an excellent review of the extensive literature on formaldehyde photochemistry see ref 3) have attracted much interest, in particular in the framework of the theory of radiationless transitions. These experiments indicate that photochemical decomposition near the threshold most probably proceeds by a sequential mechanism



In order to clarify whether $\text{S}_1 \rightarrow \text{S}_0^*$ or $\text{S}_0^* \rightarrow$ products are the "rate-determining processes", the theory of nonradiative processes was applied³ to the coupling of single vibronic states in S_1 and the "lumpy continuum" of S_0^* . Furthermore, the dynamics of unimolecular dissociation of S_0^* was investigated in a series of model calculations: the S_0 potential energy surface was calculated ab initio;⁴ anharmonic vibrational densities of states were de-

termined;⁵ the effect of tunneling was studied in detail;^{6,7} for $J = 0$ mode-selective dissociation was postulated;^{8,9} classical trajectory calculations were performed in ref 10.

There have been several RRKM type statistical calculations of the specific rate constants for the molecular elimination channel (eq 1) of the dissociation (see ref 6 and other work cited in ref 3). The simple bond fission channel (eq 2), however, has received much less attention. So far, these calculations do not appear to provide a sufficiently detailed basis for application to thermal and photochemical dissociation experiments. In both cases, the J dependence of the rate constants is of crucial importance and has to be accounted for explicitly. For the molecular elimination channel, the contributions of the K rotor (in symmetrical top representation) to the rovibrational density of states $\rho(E, J)$, as well as to the rovibrational number of activated complex states $W(E, J)$, have to be implemented into a RRKM calculation together with the J dependence of the threshold energy. To our knowledge no explicit calculations of this type for the specific rate constant

$$k(E, J) = \frac{W(E, J)}{h\rho(E, J)} \quad (4)$$

have been published so far, except for $J = 0$. The treatment of the J dependence of the specific rate constants $k(E, J)$ for the simple bond fission is much more difficult since complicated angular momentum coupling restrictions have to be accounted for. Based on the detailed statistical adiabatic channel model,¹¹ we have recently formulated a simplified adiabatic channel model^{12,13} which allows for a convenient handling of the angular momentum problem. This model includes all elements of loose phase space theory, but in addition it does not neglect the important angular parts of the interaction potential between the

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dissociation fragments. This simplified statistical adiabatic channel model is applied in the present work to the simple bond fission channel (eq 2) whereas an RRKM type calculation including rotational contributions is applied to the molecular elimination channel (eq 1). Finally, the contribution of tunneling to $k(E, J)$ for the molecular elimination channel (eq 1) is evaluated following the treatment of ref 6 and 7.

Our calculations can be implemented in the framework of the theory of two-channel thermal unimolecular reactions.¹⁴⁻¹⁶ The formaldehyde system shows the complication recognized in ref 14 of a "channel switching" for rotating molecules: with increasing J the energetic order of the threshold energies for the two channels changes. Our present calculation provides a first explicit demonstration of this effect. The consequences on the low- and high-pressure range of the thermal decomposition of formaldehyde will be illustrated in ref 17. (A first attempt was made in ref 18 to apply unimolecular rate theory including tunneling to the thermal decomposition. However, the effects of rotation discussed in the present work and the effects of weak collisions discussed in ref 14, which are of crucial importance for the present system, were neglected. Therefore, a more realistic treatment¹⁷ is required.) Our calculations also apply to multiphoton IR excitation of formaldehyde¹⁹⁻²² proceeding on the electronic ground state. It appears uncertain to what extent the specific rate constants $k(E, J)$ for the electronic ground-state explicitly enter into the properties of photochemical decomposition or fluorescence lifetimes. Our calculations show that, except in the tunneling range,^{6,7} fluorescence decay rate constants are always much smaller than the calculated $k(E, J)$ for dissociation. This suggests a "rate-determining" $S_1 \rightarrow S_0^*$ radiationless process governing photochemistry. Nevertheless, the $S_1 \rightarrow S_0^*$ coupling will depend also on the dissociative widths of the S_0^* states as derived from our $k(E, J)$ calculations.

Specific Rate Constants $k(E, J)$ for $H_2CO \rightarrow H_2 + CO$

The molecular elimination channel (eq 1) is characterized by a well-localized transition state. Model potential energy surfaces²³ and ab initio calculations^{4,6,7} allow for a reliable prediction of the corresponding activated complex parameters. We used the scaled frequencies of ref 7 for H_2CO ; for D_2CO we have scaled the frequencies of ref 4 and 6 by a factor corresponding to the difference of the H_2CO values of ref 6 and 7. The molecular frequencies of H_2CO and D_2CO were taken as the "harmonized" values of ref 24, since explicit anharmonicity corrections are applied later on. A preliminary value of the threshold energy of $E_0 \approx 83.5 \text{ kcal mol}^{-1} \approx 29205 \text{ cm}^{-1}$ is used for H_2CO . A fine adjustment of this value may become necessary when more is known. The corresponding value for D_2CO with the different zero-point energies follows as $E_0 \approx 84.7 \text{ kcal mol}^{-1} \approx 29620 \text{ cm}^{-1}$. Moments of inertia for the activated complex geometries of ref 4 (slightly scaled) are given with the frequency parameters in Table I.

Specific rate constants $k(E, J)$ with these molecular parameters are calculated in the framework of RRKM theory. $W(E, J)$ is given by^{6,11,13}

$$W(E, J) \approx \sum_{K=J}^J W^{\ddagger}(E - E_0(J) - (A_0^{\ddagger} - B^{\ddagger})K^2) \quad (5)$$

$B^{\ddagger}$ denotes $(B_0^{\ddagger} + C_0^{\ddagger})/2$, similarly we use $B = (B_0 + C_0)/2$. W is obtained in the harmonic oscillator approximation by exact

TABLE I: Molecular Parameters (in cm^{-1}) of H_2CO and D_2CO

	H_2CO	D_2CO		H_2CO	D_2CO
ω_1	2944 ^a	2144 ^a	A_0	9.405 ^d	4.702 ^d
ω_2	1764	1717	B_0	1.295	1.074
ω_3	1563	1140	C_0	1.134	0.872
ω_4	1191	955	A_0^*	7.555 ^e	3.958 ^e
ω_5	3009	2255	B_0^*	1.230	0.962
ω_6	1288	1014	C_0^*	1.055	0.771
ω_1^*	3125 ^b	2475 ^c	σ	2	2
ω_2^*	1830	1663	$E_0(1)$	29200	29630 ^f
ω_3^*	1523	1098	$E_0(2)$	30080	30920 ^f
ω_4^*	839	608			
ω_5^*	2026i	1682i			
ω_6^*	936	720			

^a "Harmonized" frequencies of ref 24. ^b Scaled ab initio calculations of ref 7. ^c Ab initio calculations of ref 4 given in ref 6, scaled according to the H_2CO difference between ref 6 and 7. ^d For experimental configuration used in ref 4. ^e For ab initio calculation of ref 4, scaled according to difference between ab initio calculation and experimental configuration of H_2CO . ^f From H_2CO values accounting for different zero-point energies.

counting with the Beyer-Swinehart algorithm.²⁵ K is constrained by $|K| \leq J$ and by the condition of a positive argument of W . The effective threshold energy

$$E_0(J) = E_0 + (B^{\ddagger} - B)J(J+1) \quad (6)$$

differs only slightly from $E_0 \equiv E_0(J=0)$. Note that throughout this paper E is counted from the zero-point energy level of the molecule at $J = 0$. Rovibrational densities of states

$$\rho(E, J) \approx \sum_{K=J}^J \rho_{\text{vib}}(E - (A_0 - B)K^2) \quad (7)$$

are calculated with analogous K constraints by the Beyer-Swinehart algorithm for harmonic oscillators. They are divided by the symmetry number $\rho = 2$. (The additional factor 2 for planarity of the equilibrium and transition state configurations is omitted in W and ρ .) It is difficult to apply adequate anharmonicity corrections. Different from the policies of ref 5, we use a model of coupled Morse oscillators for which, to a good approximation, an anharmonicity factor $F_{\text{anh}}(E, J)$ of

$$F_{\text{anh}}(E, J) \approx \prod_{i=1}^s \left(1 + \frac{E_i' - BJ(J+1)}{(2s-3)(D_i - BJ(J+1))} \right) \quad (8)$$

has been derived¹³ (with $s = 6$ oscillators, $E_i' = E + \omega_i/2$, $D_i = E_{0i} + \omega_i/2$, and E_{0i} the estimated dissociation energies of oscillator i , we use $E_{0i} = E_0$ except for $D_2 \approx 43000 \text{ cm}^{-1}$).

Including the K rotor in the calculation of ρ and W is justified by the rapid rotational and vibrational mixing at $E > 4000 \text{ cm}^{-1}$ observed in ref 26. The calculation for $k(E, J)$ following eq 4-8 today is trivial. Nevertheless, apparently no explicit representations of the J dependence for an elimination reaction of type 1 have been published so far. For this reason we show our results in detail in Table II and Figure 1 for H_2CO , and in Table III and Figure 2 for D_2CO . (Some results for $J > 0$ published in ref 6 are in agreement with the present calculation.) For $J > 0$, in Figure 1 one observes an increase of the threshold energies. Because of the very similar values of rotational constants $B^{\ddagger}$ and B , this shift is nearly equal to the rotational energy $BJ(J+1)$ such that the "effective threshold for vibrational energy" $E_0 + B^{\ddagger}J(J+1) - BJ(J+1)$ nearly does not change. The threshold rate constants $k(E_0 + B^{\ddagger}J(J+1), J)$ drop markedly below the value for $J = 0$ because of the increasing contribution of the K states to $\rho(E, J)$. At energies not too far above the threshold energies, the K contribution to $W(E, J)$ compensates the K contribution in $\rho(E, J)$ such that the $J > 0$ curves become just shifted against the $J = 0$ curve by roughly the amount $BJ(J+1)$. In reality this shift is slightly

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TABLE II: Specific Rate Constants $k(E, J)$ for $\text{H}_2\text{CO} \rightarrow \text{H}_2 + \text{CO}^a$

E/cm^{-1}	$J = 0$	$J = 5$	$J = 10$	$J = 20$	$J = 30$	$J = 50$
29500	2.6×10^9	2.8×10^9	1.5×10^9			
30000	2.4×10^9	2.5×10^9	2.7×10^9	1.1×10^9		
30500	6.9×10^9	7.1×10^9	5.8×10^9	1.8×10^9	7.8×10^8	
31000	1.3×10^{10}	1.1×10^{10}	8.5×10^9	4.2×10^9	1.3×10^9	
31500	1.6×10^{10}	1.6×10^{10}	1.5×10^{10}	7.0×10^9	3.2×10^9	
32000	2.8×10^{10}	2.6×10^{10}	2.0×10^{10}	1.2×10^{10}	5.1×10^9	
32500	3.4×10^{10}	3.2×10^{10}	2.9×10^{10}	1.8×10^{10}	8.9×10^9	8.5×10^8
33000	4.8×10^{10}	4.7×10^{10}	4.0×10^{10}	2.5×10^{10}	1.3×10^{10}	1.6×10^9
33500	6.1×10^{10}	5.8×10^{10}	5.3×10^{10}	3.5×10^{10}	1.9×10^{10}	3.1×10^9
34000	7.6×10^{10}	7.5×10^{10}	6.8×10^{10}	4.6×10^{10}	2.7×10^{10}	5.7×10^9
34500	9.7×10^{10}	9.5×10^{10}	8.6×10^{10}	6.0×10^{10}	3.6×10^{10}	8.5×10^9
35000	1.1×10^{11}	1.1×10^{11}	1.0×10^{11}	7.8×10^{10}	4.8×10^{10}	1.3×10^{10}
35500	1.4×10^{11}	1.4×10^{11}	1.3×10^{11}	9.7×10^{10}	6.2×10^{10}	1.8×10^{10}
36000	1.7×10^{11}	1.7×10^{11}	1.6×10^{11}	1.2×10^{11}	7.9×10^{10}	2.6×10^{10}
36500	2.0×10^{11}	2.0×10^{11}	1.9×10^{11}	1.5×10^{11}	9.8×10^{10}	3.4×10^{10}
37000	2.3×10^{11}	2.3×10^{11}	2.2×10^{11}	1.8×10^{11}	1.2×10^{11}	4.5×10^{10}
37500	2.7×10^{11}	2.7×10^{11}	2.5×10^{11}	2.1×10^{11}	1.5×10^{11}	5.8×10^{10}
38000	3.1×10^{11}	3.1×10^{11}	2.9×10^{11}	2.4×10^{11}	1.8×10^{11}	7.3×10^{10}
38500	3.5×10^{11}	3.5×10^{11}	3.3×10^{11}	2.8×10^{11}	2.1×10^{11}	9.0×10^{10}
39000	4.0×10^{11}	4.0×10^{11}	3.8×10^{11}	3.2×10^{11}	2.4×10^{11}	1.1×10^{11}
$E_0(J)/\text{cm}^{-1}$	29200	29230	29330	29680	30260	32110
$k(E_0)/\text{s}^{-1}$	2.8×10^9	4.7×10^8	2.7×10^8	1.6×10^8	1.3×10^8	1.1×10^8

^a Values in s^{-1} ; threshold rate constants (omitting factor of 2 symmetry number for planarity) are given at the bottom of the table.

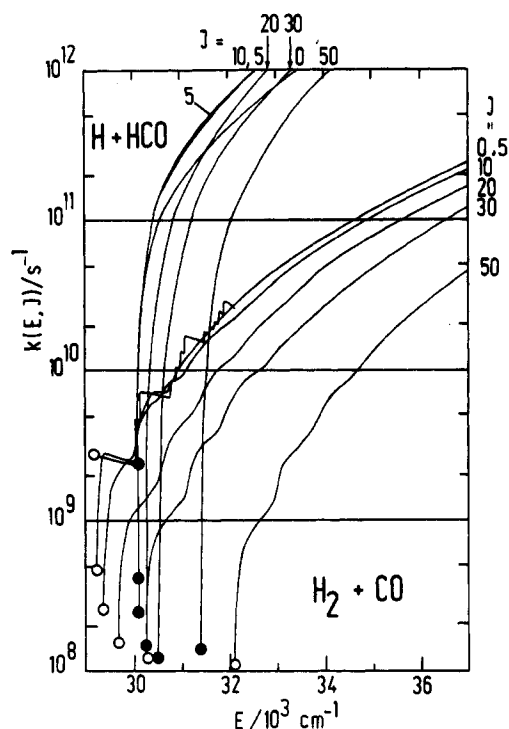


Figure 1. Specific rate constants $k(E, J)$ for the unimolecular elimination $\text{H}_2\text{CO} \rightarrow \text{H}_2 + \text{CO}$ and the unimolecular bond fission $\text{H}_2\text{CO} \rightarrow \text{H} + \text{HCO}$. (Threshold rate constants given by $\circ$ and $\bullet$; the step structure of the $J = 0$ curve for $\text{H}_2\text{CO} \rightarrow \text{H}_2 + \text{CO}$ becomes smooth at $E > 32000 \text{ cm}^{-1}$; tunneling contributions are included in Figure 3.)

larger because the constraint $E - E(J) \geq (A_0^* - B^*)K^2$ in $W(E, J)$ operates earlier than the constraint $E > (A_c - B)K^2$ in $\rho(E, J)$. For $J > 0$, the K contribution smoothens the marked step structure of the $k(E, J=0)$ curve shown in Figure 1. Nevertheless, some smoothed step structure remains visible even at $J = 50$.

The D_2CO results are quite similar to those for H_2CO . However, the larger density of states because of smaller vibrational frequencies produces smaller threshold rate constants as illustrated in Figure 2.

Specific Rate Constants $k(E, J)$ for $\text{H}_2\text{CO} \rightarrow \text{H} + \text{HCO}$

The simple bond fission channel (eq 2) involves a Morse-type potential along the reaction coordinate q of a H-C bond. This

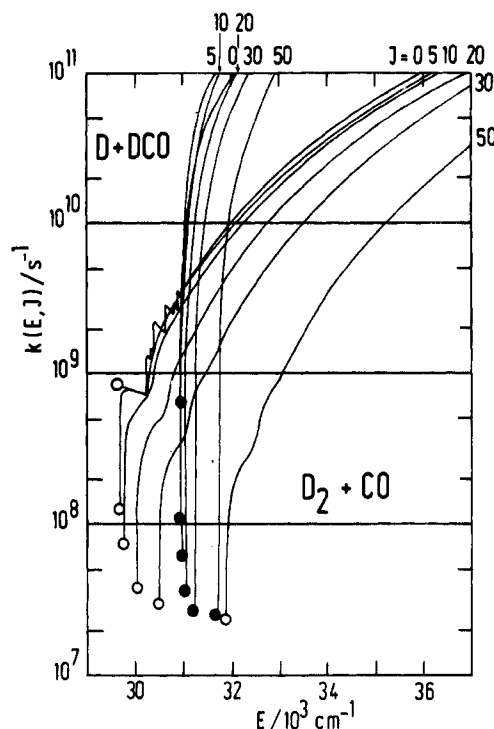


Figure 2. Specific rate constants $k(E, J)$ for $\text{D}_2\text{CO} \rightarrow \text{D}_2 + \text{CO}$ and $\text{D}_2\text{CO} \rightarrow \text{D} + \text{DCO}$ (remarks as in Figure 1).

reaction is accompanied by the transformation of two oscillators into fragment rotations. This transition is governed by coupling with the overall rotation of the molecule. In the simplified statistical adiabatic channel model of ref 12 and 13, the corresponding coordinates of intermediate character between oscillators and rotors are factorized. Their numbers of states are used in the starting array of a Beyer-Swinehart counting algorithm applied to the other harmonic oscillator type coordinates.²⁷ The factorization introduces decoupling errors which are corrected for by overall angular momentum coupling factors $F_{\text{AM}}(E, J)$. The centrifugal barriers $E_0(J)$ are obtained from the q dependences of the Morse potential, the rotational energy, and the vibrational

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TABLE III: $k(E, J)$ for $D_2CO \rightarrow D_2 + CO^a$

E/cm^{-1}	$J = 0$	$J = 5$	$J = 10$	$J = 20$	$J = 30$	$J = 50$
30000	7.8×10^8	7.4×10^8	6.8×10^8	8.0×10^8		
30500	2.2×10^9	2.1×10^9	1.4×10^9	4.9×10^8	1.5×10^8	
31000	3.8×10^9	3.5×10^9	2.7×10^9	1.4×10^9	3.9×10^8	
31500	6.5×10^9	5.8×10^9	4.8×10^9	2.9×10^9	1.1×10^9	
32000	9.8×10^9	8.9×10^9	7.8×10^9	5.0×10^9	2.3×10^9	1.8×10^8
32500	1.5×10^{10}	1.3×10^{10}	1.2×10^{10}	8.1×10^9	4.2×10^9	4.2×10^8
33000	2.2×10^{10}	1.9×10^{10}	1.8×10^{10}	1.2×10^{10}	6.9×10^9	9.6×10^8
33500	2.9×10^{10}	2.7×10^{10}	2.5×10^{10}	1.8×10^{10}	1.0×10^{10}	2.0×10^9
34000	4.0×10^{10}	3.6×10^{10}	3.4×10^{10}	2.5×10^{10}	1.5×10^{10}	3.5×10^9
34500	5.1×10^{10}	4.7×10^{10}	4.5×10^{10}	3.4×10^{10}	2.2×10^{10}	5.8×10^9
35000	6.5×10^{10}	5.9×10^{10}	5.7×10^{10}	4.4×10^{10}	3.0×10^{10}	8.8×10^9
35500	8.2×10^{10}	7.5×10^{10}	7.2×10^{10}	5.9×10^{10}	4.0×10^{10}	1.3×10^{10}
36000	1.0×10^{11}	9.2×10^{10}	8.8×10^{10}	7.1×10^{10}	5.2×10^{10}	1.8×10^{10}
36500	1.2×10^{11}	1.1×10^{11}	1.1×10^{11}	8.8×10^{10}	6.6×10^{10}	2.4×10^{10}
37000	1.4×10^{11}	1.3×10^{11}	1.3×10^{11}	1.1×10^{11}	8.2×10^{10}	3.3×10^{10}
37500	1.7×10^{11}	1.6×10^{11}	1.5×10^{11}	1.3×10^{11}	1.0×10^{11}	4.3×10^{10}
38000	2.0×10^{11}	1.9×10^{11}	1.8×10^{11}	1.5×10^{11}	1.2×10^{11}	5.4×10^{10}
38500	2.3×10^{11}	2.1×10^{11}	2.1×10^{11}	1.7×10^{11}	1.4×10^{11}	6.8×10^{10}
39000	2.7×10^{11}	2.5×10^{11}	2.4×10^{11}	2.0×10^{11}	1.7×10^{11}	8.4×10^{10}
39500	3.0×10^{11}	2.8×10^{11}	2.8×10^{11}	2.4×10^{11}	2.0×10^{11}	1.0×10^{11}
$E_0(J)/\text{cm}^{-1}$	29630	29660	29730	29990	30440	31840
$k(E_0)/\text{s}^{-1}$	8.5×10^8	1.2×10^8	7.5×10^7	3.8×10^7	3.0×10^7	2.4×10^7

^a See footnote a to Table II.TABLE IV: $k(E, J)$ for $H_2CO \rightarrow H + HCO^a$

E/cm^{-1}	$J = 0$	$J = 5$	$J = 10$	$J = 20$	$J = 30$	$J = 50$
30100	6.8×10^9	1.0×10^9				
30200	2.6×10^{10}	2.4×10^{10}				
30300	4.6×10^{10}	5.8×10^{10}	9.7×10^9			
30400	6.5×10^{10}	9.1×10^{10}	5.5×10^{10}	2.4×10^9		
30500	8.5×10^{10}	1.2×10^{11}	8.3×10^{10}	1.3×10^{10}		
30600	1.0×10^{11}	1.5×10^{11}	1.1×10^{11}	2.9×10^{10}	1.2×10^8	
30700	1.2×10^{11}	1.8×10^{11}	1.5×10^{11}	4.7×10^{10}	7.1×10^9	
30800	1.4×10^{11}	2.1×10^{11}	1.8×10^{11}	6.7×10^{10}	1.9×10^{10}	
30900	1.6×10^{11}	2.4×10^{11}	2.2×10^{11}	8.8×10^{10}	3.4×10^{10}	
31000	1.7×10^{11}	2.7×10^{11}	2.5×10^{11}	1.1×10^{11}	5.1×10^{10}	
31500	2.7×10^{11}	4.4×10^{11}	4.2×10^{11}	1.4×10^{11}	6.9×10^{10}	
32000	4.3×10^{11}	6.5×10^{11}	6.4×10^{11}	2.7×10^{11}	1.7×10^{11}	8.3×10^9
32500	5.6×10^{11}	9.3×10^{11}	9.1×10^{11}	4.8×10^{11}	3.3×10^{11}	9.4×10^{10}
33000	7.7×10^{11}	1.3×10^{12}	1.3×10^{12}	7.5×10^{11}	5.3×10^{11}	2.0×10^{11}
33500	1.0×10^{12}	1.7×10^{12}	1.7×10^{12}	1.1×10^{12}	8.1×10^{11}	3.8×10^{11}
34000	1.3×10^{12}	2.2×10^{12}	2.1×10^{12}	1.6×10^{12}	1.2×10^{12}	6.0×10^{11}
34500	1.6×10^{12}	2.7×10^{12}	2.7×10^{12}	2.1×10^{12}	1.6×10^{12}	9.2×10^{11}
35000	1.9×10^{12}	3.3×10^{12}	3.3×10^{12}	2.6×10^{12}	2.2×10^{12}	1.3×10^{12}
35500	2.3×10^{12}	3.9×10^{12}	3.9×10^{12}	3.2×10^{12}	2.8×10^{12}	1.8×10^{12}
36000	2.7×10^{12}	4.7×10^{12}	4.6×10^{12}	3.9×10^{12}	3.5×10^{12}	2.3×10^{12}
36500	3.2×10^{12}	5.5×10^{12}	5.4×10^{12}	4.6×10^{12}	4.3×10^{12}	3.0×10^{12}
37000	3.6×10^{12}	6.2×10^{12}	6.2×10^{12}	5.4×10^{12}	5.2×10^{12}	3.7×10^{12}
37500	4.1×10^{12}	7.1×10^{12}	7.1×10^{12}	6.2×10^{12}	6.2×10^{12}	4.5×10^{12}
38000	4.6×10^{12}	8.0×10^{12}	8.0×10^{12}	7.1×10^{12}	7.3×10^{12}	5.4×10^{12}
38500	5.2×10^{12}	9.0×10^{12}	9.0×10^{12}	8.0×10^{12}	8.4×10^{12}	6.4×10^{12}
39000	5.7×10^{12}	9.9×10^{12}	1.0×10^{13}	9.0×10^{12}	9.4×10^{12}	7.5×10^{12}
39500	6.3×10^{12}	1.1×10^{13}	1.1×10^{13}	1.0×10^{13}	1.1×10^{13}	8.6×10^{12}
40000	6.9×10^{12}	1.2×10^{13}	1.2×10^{13}	1.1×10^{13}	1.2×10^{13}	9.8×10^{12}
				1.2×10^{13}	1.3×10^{13}	1.1×10^{13}
$E_0(J)/\text{cm}^{-1}$	30080	30100	30130	30260	30500	31390
$k(E_0)/\text{s}^{-1}$	2.5×10^9	4.2×10^8	2.4×10^8	1.4×10^8	1.2×10^8	1.3×10^8

^a See footnote a to Table II.

zero-point energies. For details of this approach the reader is referred to ref 12 and 13. Apart from the H_2CO parameters given in Table I, the calculation requires the ratio α/β of the looseness parameter α and the Morse parameter β . From the force constant, or from the frequency ω_5 , of the vibration in the reaction coordinate, one obtains β values near $1.94 \text{ \AA}^{-1}$. In the absence of a fit to adequate experiments, we use for α a "standard" value¹¹ of $1 \text{ \AA}^{-1}$. The dissociation energy for reaction 2 is near $E_0 = 86 \text{ kcal mol}^{-1} \approx 30080 \text{ cm}^{-1}$ (see the summary in ref 28). HCO frequencies of 2488, 1083, and 1820 cm^{-1} from ref 29 were used;

DCO frequencies of 1930, 847, and 1761 cm^{-1} from ref 29 were slightly scaled in analogy to HCO.³⁰ HCO rotational constants of 18.648, 1.523, and 1.408 cm^{-1} and DCO rotational constants of 9.648, 1.334, and 1.172 cm^{-1} follow from the preferred HCO geometry of ref 30.

The results of our $k(E, J)$ calculations are given in Tables IV and V. They are included in Figures 1 and 2 and compared with the results for the molecular elimination channels. The threshold energies $E_0(J)$ increase with increasing J much less than in the molecular elimination channels. The centrifugal energies are

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TABLE V: $k(E, J)$ for $\text{D}_2\text{CO} \rightarrow \text{D} + \text{DCO}^a$

E/cm^{-1}	$J = 0$	$J = 5$	$J = 10$	$J = 20$	$J = 30$	$J = 50$
31000	7.6×10^9	7.6×10^9	3.1×10^9			
31100	1.6×10^{10}	2.3×10^{10}	1.2×10^{10}	3.3×10^9		
31200	2.5×10^{10}	3.7×10^{10}	2.4×10^{10}	9.6×10^9	1.1×10^9	
31300	3.4×10^{10}	5.1×10^{10}	3.8×10^{10}	1.7×10^{10}	5.3×10^9	
31400	4.2×10^{10}	6.5×10^{10}	5.4×10^{10}	2.7×10^{10}	1.1×10^{10}	
31500	5.1×10^{10}	7.9×10^{10}	7.0×10^{10}	3.7×10^{10}	1.8×10^{10}	
31600	5.9×10^{10}	9.2×10^{10}	8.6×10^{10}	4.7×10^{10}	2.6×10^{10}	
31700	6.7×10^{10}	1.1×10^{11}	9.9×10^{10}	5.8×10^{10}	3.4×10^{10}	9.9×10^8
31800	7.5×10^{10}	1.2×10^{11}	1.1×10^{11}	7.0×10^{10}	4.3×10^{10}	4.7×10^9
31900	8.6×10^{11}	1.4×10^{11}	1.3×10^{11}	8.3×10^{10}	5.2×10^{10}	9.8×10^9
32000	9.8×10^{11}	1.6×10^{11}	1.5×10^{11}	9.9×10^{10}	6.2×10^{10}	1.6×10^{10}
32500	1.6×10^{11}	2.6×10^{11}	2.5×10^{11}	2.0×10^{11}	1.4×10^{11}	5.4×10^{10}
33000	2.4×10^{11}	4.0×10^{11}	3.8×10^{11}	3.3×10^{11}	2.3×10^{11}	1.2×10^{11}
33500	3.5×10^{11}	5.9×10^{11}	5.7×10^{11}	5.3×10^{11}	3.8×10^{11}	2.0×10^{11}
34000	4.9×10^{11}	8.2×10^{11}	8.1×10^{11}	7.6×10^{11}	5.8×10^{11}	3.4×10^{11}
34500	6.4×10^{11}	1.1×10^{12}	1.1×10^{12}	1.0×10^{12}	8.4×10^{11}	5.2×10^{11}
35000	8.3×10^{11}	1.4×10^{12}	1.4×10^{12}	1.3×10^{12}	1.1×10^{12}	7.4×10^{11}
35500	1.0×10^{12}	1.8×10^{12}	1.8×10^{12}	1.7×10^{12}	1.5×10^{12}	9.9×10^{11}
36000	1.3×10^{12}	2.2×10^{12}	2.2×10^{12}	2.1×10^{12}	2.0×10^{12}	1.3×10^{12}
36500	1.6×10^{12}	2.6×10^{12}	2.6×10^{12}	2.5×10^{12}	2.5×10^{12}	1.7×10^{12}
37000	1.8×10^{12}	3.1×10^{12}	3.1×10^{12}	3.0×10^{12}	3.0×10^{12}	2.2×10^{12}
37500	2.1×10^{12}	3.7×10^{12}	3.7×10^{12}	3.6×10^{12}	3.5×10^{12}	2.6×10^{12}
38000	2.5×10^{12}	4.3×10^{12}	4.3×10^{12}	4.1×10^{12}	4.1×10^{12}	3.2×10^{12}
38500	2.8×10^{12}	4.9×10^{12}	4.9×10^{12}	4.8×10^{12}	4.7×10^{12}	3.8×10^{12}
39000	3.2×10^{12}	5.5×10^{12}	5.5×10^{12}	5.4×10^{12}	5.3×10^{12}	4.4×10^{12}
39500	3.6×10^{12}	6.2×10^{12}	6.2×10^{12}	6.1×10^{12}	6.0×10^{12}	5.1×10^{12}
40000	4.0×10^{12}	6.9×10^{12}	6.9×10^{12}	6.8×10^{12}	6.6×10^{12}	5.9×10^{12}
$E_0(J)/\text{cm}^{-1}$	30920	30930	30950	31020	31150	31650
$k(E_0)/\text{s}^{-1}$	6.4×10^8	1.1×10^8	6.0×10^7	3.5×10^7	2.7×10^7	2.5×10^7

^aSee footnote a to Table II.

smaller than the corresponding rotational energies $BJ(J+1)$ such that the "effective threshold for vibrational energy" decreases markedly with increasing J . As a consequence there is a "channel switching" (near $J = 35$ for H_2CO) where the threshold energies for both channels are equal; channel 1 has a smaller threshold energy at $J < 35$, and a larger threshold energy at $J > 35$, than channel 2. Of course, the exact J value for channel switching crucially depends on the difference ΔE_0 of E_0 values for $J = 0$ which remains uncertain. However, since the branching in thermal dissociation sensitivity depends on the difference in threshold energies, these experiments provide a direct access to ΔE_0 .¹⁷

The J dependence of specific rate constants $k(E, J)$ for simple bond fission reactions differs in several details from the corresponding molecular elimination reactions. Similar to the examples of ref 13, one observes a crossing of the curves for different J values. This is due to the compensation of several factors. At first, due to the increase of $\rho(E, J)$ from K contributions, the threshold rate constants $k(E=E_0(J), J)$ decrease with increasing J . At high J , $k(E=E_0(J), J)$ increases again, because the $(2J+1)$ factor in $\rho(E, J)$ is compensated by the diminishing amount $E-BJ(J+1)$ of distributable energy in $\rho(E, J)$. The $k(E, J)$ curves are not simply shifted along the energy scale according to the threshold energies $E_0(J)$. Instead, the K contribution to $W(E, J)$ becomes increasingly important with increasing J . This contribution is characterized partly by an increase in statistical weight of maximum $2J+1$ which, however, is largely diminished by the limited amount of distributable energy. The details of this contribution are complicated due to the various angular momentum coupling effects. At energies higher than shown in Figures 1, the various curves not only cross the $J = 0$ curve, but cross each other such that the ordering is completely inverted again. Similar behavior was demonstrated for C_2H_6 in ref 13.

Tunneling Contributions

Whereas the simple bond fission (eq 2) involves only minor tunneling contributions, the specific rate constants $k(E, J)$ for the molecular elimination are markedly influenced by tunneling. Following the treatment of ref 6 and 7, we have calculated the tunneling contributions by the simple Eckart procedure. As shown in ref 7, for formaldehyde this simple approach is fully sufficient.

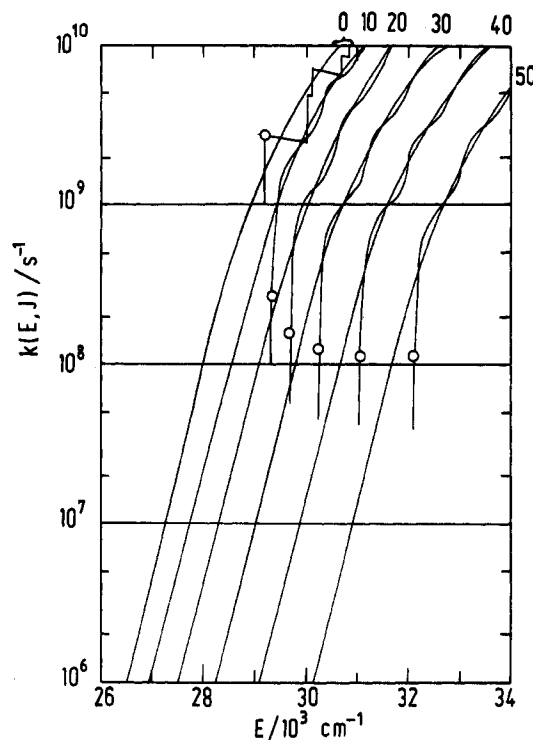


Figure 3. Specific rate constants $k(E, J)$ for the elimination reaction $\text{H}_2\text{CO} \rightarrow \text{H}_2 + \text{CO}$ (including tunneling; the results without tunneling from figure 1 are also shown).

Using the molecular parameters of Table I, we obtain specific rate constants $k(E, J)$ for the H_2CO and D_2CO elimination reactions such as listed in Table VI, and illustrated for H_2CO in Figure 3. Tunneling very efficiently smoothens the discontinuous step structure from the RRKM calculations. Tunneling also produces nonvanishing rate constants $k(E, J)$ below the "threshold energies" E_0 of the reaction. The decay of $k(E, J)$ with decreasing energy is more pronounced for D_2CO than for H_2CO . One should note

TABLE VI: Specific Rate Constants $k(E, J=0)$ for the Reactions $\text{H}_2\text{CO} \rightarrow \text{H}_2 + \text{CO}$ and $\text{D}_2\text{CO} \rightarrow \text{D}_2 + \text{CO}$ Including Tunneling Contributions

$\text{H}_2\text{CO} \rightarrow \text{H}_2 + \text{CO}$		$\text{D}_2\text{CO} \rightarrow \text{D}_2 + \text{CO}$	
E/cm^{-1}	$k(E, J=0)/\text{s}^{-1}$	E/cm^{-1}	$k(E, J=0)/\text{s}^{-1}$
24 300	9.9×10^2	24 730	4.6
25 000	9.0×10^3	25 430	8.3×10^1
25 700	8.0×10^4	26 130	1.4×10^3
26 400	7.1×10^5	26 830	2.2×10^4
27 100	6.2×10^6	27 530	3.4×10^5
27 800	5.1×10^7	28 230	5.0×10^6
28 500	3.9×10^8	28 930	6.7×10^7
29 200 ($=E_0$)	1.8×10^9	29 630 ($=E_0$)	5.5×10^8
29 900	4.7×10^9	30 430	1.8×10^9
30 600	9.6×10^9	31 130	4.5×10^9

that our rate constants include a factor of 2 for rotational symmetry of the molecule. They do not include the additional mode-specific effects due to a planar reaction path which were discussed in ref 8 and 9. Our rate constants listed in Table VI essentially agree with the results of ref 7 (Table IV, column

"scaled", corrected by a factor of 2 for symmetry) for H_2CO at $J = 0$. Our rate constants are about a factor of 1.3 lower than the results of ref 7. This discrepancy is due to the anharmonicity contribution included here, but neglected in ref 7.

Conclusions

Not unexpectedly the calculations show simple bond fission to be dominant except very close to or below the threshold energy of this channel. The specific rate constants are markedly larger than fluorescence decay rate constants.³ The J dependence of the rate constants is complicated showing channel switching. This behavior poses particularly problems in the analysis of photolysis product yields and IR multiphoton and thermal dissociation branching ratios. Further analysis of these phenomena should be based on $k(E, J)$ calculations of the present kind.

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Registry No. H_2CO , 50-00-0; D_2CO , 1664-98-8.

Photochemical Debromination of Meso-Substituted Bromoanthracenes Studied by Steady-State Photolysis and Laser Photolysis

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Debrominations of 9-bromoanthracene (BA) and 9,10-dibromoanthracene (DBA) in acetonitrile containing triethylamine (TEA) or *N,N*-dimethylaniline (DMA) have been studied by means of steady-state photolysis and laser photolysis. By the addition of TEA, the decay constants of the lowest excited singlet states of BA and DBA increase and the maximum yields of the triplet states decrease. The singlet quenching rate constants by TEA are calculated to be of the order of $10^{10} \text{ M}^{-1} \text{ s}^{-1}$, showing that the reactions are diffusion controlled. Compared with the result of γ -radiolysis and pulse radiolysis, it is suggested that the photochemical debrominations of BA and DBA in the presence of amines take place via the anion radicals which are produced through exciplexes between amines and the lowest excited singlet states of bromoanthracenes.

Introduction

The nonradiative processes of the fluorescing states in anthracene derivatives have been studied extensively,^{1,2} and it is well-known that the fluorescence quantum yields of 9- and 9,10-substituted anthracenes in fluid media decrease as the temperature is increased. This temperature dependence has been attributed to thermally activated intersystem crossing (isc) from the lowest excited singlet state S_1 to an adjacent higher excited triplet state T_n .

Our direct measurements of the temperature dependence of the fluorescence lifetimes of 9-bromoanthracene (BA) and 9,10-dibromoanthracene (DBA)³ have allowed the determination of the rate constant for isc in the form $k_{\text{isc}} = A_{\text{isc}} \exp(-\Delta E/kT)$, with activation energies of 600 and 1100 cm^{-1} for BA and DBA in 3-methylpentane, respectively. We have also reported that the solvent dependence of the nonradiative transition rates of the lowest excited singlet states in BA and DBA⁴ is due to the different extent

to which S_1 and T_n are stabilized by the solvent, in accordance with the result of Wu and Ware for DBA.⁵

On the other hand, it is well-known that photoinduced dehalogenations of halogenated aromatic compounds are accelerated by the addition of several amines. And it has been suggested that radical anions of halogenated compounds which are produced through exciplexes are the reaction intermediates which break down to give aryl radicals and halogen anion.⁶⁻¹²

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